EquaLive

Glucosamine 750mg Capsule

Each capsule contains

Glucosamine sulfate potassium	
Chloride	996 mg
Equiv. to Glucosamine sulfate	750 mg

Description

A dark blue OP./ w hite OP. size '00' hard vegetable capsule.

Pharmacodynamics

The body manufactures glucosamine — a molecule made up of glucose and an amine. Glucosamine stimulates the manufacturer of glycosaminoglycans, which are key structure materials of cartilage.

The sulfate salt of glucosamine forms half of the disaccharide subunit of karatan sulfate, which is decrease in osteoarthtitis, and of hyaluronic acid, which is found in both articular cartilage and synovial fluid.

Glucosamine sulfate is used as an adjunct in the treatment of osteoarthritis or degenerative joint disease. Glucosamine enhances cartilage proteoglycan synthesis, thereby inhibiting deterioration of cartilage brought about by osteoarthritis and helping to maintain an equilibrium between cartilage catabolic and anabolic processes.

Numerous double-blind placebo controlled studies have shown glucosamine sulfate is effective in pain relief although it does not has a direct anti-inflammatory or analgesic effect. It appears, therefore, that the body uses glucosamine sulfate to repair damaged tissues.

Pharmacokinetics

The glucosamine component of Glucosamine sulfate is quickly and almost completely absorbed from the gastrointestinal tract following an oral dose.

Glucosamine is a small molecule (m.w. = 179) and is very soluble in water. Because of its small molecular weight and its pKa, it is well absorbed in the intestine.

In humans, about 90% of glucosamine. administered as an oral dose of Glucosamine sulfate, is absorbed. Evidence indicates absorption of glucosamine by intestinal cells is carrier mediated resulting in the active transport of glucosamine into these cells. Its acetylated derivative NAG appears to be absorbed without deacetylation of the molecule; however, this process occurs by diffusion.

After an oral dose, glucosamine concentrates in the liver, where it is either incorporated into plasma proteins, degraded into smaller molecules, or utilized for other biosynthetic process. Although absorption is very high, a substantial quantity of the absorbed glucosamine is probably modified or degraded to smaller compounds, such as water (H20), carbon dioxide (C02) and urea, it makes its "first pass" through the liver.

Glucosamine is rapidly incorporated into raticular cartilage following oral administration. In fact, articular cartilage concentrates glucosamine to a greater extent than any other structural tissue.

Elimination of glucosamine is primarily in the urine, about 11% of glucosamine or its derivatives eliminated in the feces. Part of a dose of glucosamine is eliminated as carbon dioxide via expired air.

Indication

As an adjunct for the treatment of osteoarthritis.

Dosage

Adult: 1 capsule twice a day after meal.

Interactions with other medications

Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent's effectiveness. Glucosamine is likely safe in patients with well-controlled diabetes (HbA Ic less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbAlc levels or those taking insulin, monitor blood glucose levels closely.

Individual taking diuretics may need to increase the dosage of glucosamine by one or two capsules per day.

Symptoms and treatment of overdose

High dose of glucosamine sulfate may give rise to gastric imitation. Taking large volume of fluid will help to reduce the symptoms.

Contraindication

Contraindicated in patients allergic to shellfish and who are hypersentitive to glucosamine or any of it components.

Diabetics should use cautiously as glucosamine may effect insulin sensitivity or glucose tolerance.

Side Effects

Sides effect are uncommon, when they do appear, are generally limited to:

Cardiovascular

Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

Central nervous system

Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).

Gastrointestinal

Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation. heartbum and anorexia have been described rarely during oral therapy with glucosamine.

Skin

Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

Pregnancy and lactation

Although glucosamine sulfate is safe, its use in pregnancy as with all general precautions when taking medication is best avoided.

Warning and precaution

The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.

Glucosamine is not meant to replace NSAIDs in cases of intense pain, as this is a causal therapy. It is advisable to take an anti-inflammatory drug in the initial treatment of osteoarthritis together with Glucosamine.
Caution is advised for diabetic patients and patients with allergy to shellfish and shellfish products.

Derived from seafood.

Storage condition

Store in a dry place below 30˚C. Keep out of reach of children.

Packing sizes

10 capsules
15 capsules
30 capsules
45 capsules
60 capsules

Manufacturer

Herbal Science Sdn Bhd
21, Jalan PJS 3/1, (Jalan Medan 1),
Taman Medan, 46000 Petaling Jaya,
Selangor D.E.